Dated:2nd May 2026 DAB-II Section: Fintech 4B

Scatter plot:

A scatter plot (also called a scatter diagram or scatter graph) is a type of data visualization that displays the relationship between two numerical variables by plotting individual data points on a two-dimensional graph. Eg Sale depend on Adver

- X-axis (Horizontal) → Independent variable (predictor)
- Y-axis (Vertical) → Dependent variable (outcome)
- Each dot represents one observation/data point

Some example of relation through scatter plot:

Regression Analysis

Regression is a statistical method used to model the relationship between a dependent variable (outcome) and one or more independent variables (predictors). It helps in predicting, forecasting, and understanding how variables influence each other.

Sale= 10+2adver

Sale=10+2(0)=10

Sale= 10+2(1)=12

Sale=10+2(2)=14

1. Simple Linear Regression

Models the relationship between one independent variable (X) and one dependent variable (Y).

Population Model: $Y = \alpha + \beta X + \varepsilon \Rightarrow Y = mx + c$

Term	Meaning
α	Intercept (value of Y when X = 0)
β	Slope (change in Y per unit change in X, regression coefficient)
E	Error term

Example: Predicting salary (Y) based on years of experience (X).

Sample model: $\hat{y} = a + bx$

Where

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}$$

$$a = \bar{y} - b\bar{x}$$

$\varepsilon = Y - \hat{y} \Rightarrow$ Observed-Predicted used in Hypothesis testing

Steps of performing Simple linear regression:

Step 1: Calculate the following sums, statistics & coefficients directly from calculator

Summation: $n, \sum X, \sum Y, \sum X^2, \sum Y^2, \sum XY,$

Statistics: $\bar{X}, \bar{Y}, S_x^2, S_y^2$

Coefficients: $a, b \,\&\, r$

Step 2: Write the formula of a & b, substitute the value, directly write ans, then fit the coefficient & variable in the model

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}$$

$$a = \bar{y} - b\bar{x}$$

$$\hat{y} = a + bx$$

Step:3 interpret the result

Question 1

A retail chain wants to analyze the relationship between customer footfall and daily sales. The following data were collected for 10 days:

Footfall (in hundreds)X	15	18	20	22	25	27	28	30	32	35
Sales (in thousands)Y	120	140	160	180	210	230	250	270	300	320

a) Fit the regression model sale on Footfall also interpret the result.
b) Predict daily sales when footfall is 24.
   Mode➔ 3)Stat➔ 2)a+bx➔ data enter
   Sum➔ shift+1+3
   Stat➔ Shift+1+4
   Regres➔ Shift+1+5

Step 1: Calculate the following sums, statistics & coefficients directly from calculator

Summation:

Sum	252	2180	6720	516800	58840

$$n = 10, \sum X = 252, \sum Y = 2180, \sum X^2 = 6720, \sum Y^2 = 516800, \sum XY = 58840$$

Statistics:

xbar	25.2	S2x	41.06667
ybar	218	S2Y	4617.778
a	4.721848	b	0.093936
r	0.99611		

$\bar{X} = 25.2, \bar{Y} = 218 S_x^2 = 41.0667, S_y^2 = 4617.7778$

Coefficients: $a = -48.1818, b = 10.5628 \,\&\, r = 0.99611$

Step 2: Write the formula of a & b , substitute the value , directly write ans, then fit the coefficient & variable in the model

$$n = 10, \sum X = 252, \sum Y = 2180, \sum X^2 = 6720, \sum Y^2 = 516800, \sum XY = 58840$$

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = \frac{(10 * 58840) - (252 * 2180)}{(10 * 6720) - 252^2} = 10.5628$$

$$a = \bar{y} - b\bar{x} = 218 - (10.5628 * 25.2) = -48.1818$$

$$\hat{y} = a + bx$$

Sale=-48.1818+10.5628 footfall

Step:3 interpret the result Alpha: sale will remain -48.1818 without consideration of footfall, sale will increase 10.5628 units increase in a unit of footfall

Predict daily sales when footfall is 24.==> Sale=-48.1818+10.5628 (24) =205.3254

Question 2

A manufacturer wants to investigate the relationship between **production cost** and **profit earned**. The data for 10 weeks are:

Production Cost (in thousands) X	80	85	90	95	100	105	110	115	120	125
Profit (in thousands) Y	150	165	180	195	210	225	240	260	275	290

a) Fit the regression model Profit on Production cost also interpret the result.
b) Predict the profit when production cost is 92.

Step 1: Calculate the following sums, statistics & coefficients directly from calculator

Summation:

Sum	1025	2190	107125	499800	230925

$n = 10, \sum X = 1025, \sum Y = 2190, \sum X^2 = 107125, \sum Y^2 499800, \sum XY = 230925$

Statistics:

Xbar	102.5	S2x	229.1667
Ybar	219	S2Y	2243.333

$\bar{X} = 102.5, \overline{Y = 219}\, S_x^2 = 229.1667, S_y^2 = 2243.3333$

Coefficients: $a = -101.5454, b = 3.1272 \;\&\; r = 0.999$

Step 2: Write the formula of a & b , substitute the value , directly write ans, then fit the coefficient & variable in the model

$$n = 10, \sum X = 1025, \sum Y = 2190, \sum X^2 = 107125, \sum Y^2 499800, \sum XY = 230925$$

$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = \frac{(10*230925)-(2190*1025)}{(10*107125)-(1025^2)} = 3.1272$

$a = \bar{y} - b\bar{x}$=219-3.1272*102.5=-101.5454

$\hat{y} = a + bx$ ➔ Profit= -101.5454+3.1272 Production cost

Step:3 interpret the result : Profit will -101.5454 without consideration of production cost, Profit will increase 3.1272 units increase in a unit production cost.

Production yield= 10+15 seed

Regression Coefficient Significance

1) First calculate Model standard deviation: $S_{yx} = \sqrt{\frac{\sum Y^2 - a\sum Y - b\sum XY}{n-2}}$
2) Then b coefficient standard deviation: $S_b = \frac{S_{yx}}{\sqrt{\sum(X-\bar{X})^2}} ==> \frac{S_{yx}}{\sqrt{(n-1)S_x^2}} ==> \frac{S_{yx}}{\sqrt{n*\sigma_x^2}}$

   $S_x^2 = \frac{\sum(X-\bar{X})^2}{n-1}, \sigma^2 = \frac{\sum(X-\bar{X})^2}{n}$
3) Then compute confidence interval & do hypothesis testing;

$$b \pm t_{\frac{\alpha}{2}, n-2} S_b$$

   Hypothesis Testing
   Ho The regression coefficient is insignificant (β=0)
   H1 The regression coefficient is significant (β≠0)
   Level of significance: α =
   Test Statistics: $t = \frac{b}{S_b}$
   Critical region: $t > t_{\alpha/2, n-2}$ or $t < -t_{\alpha/2, n-2}$
   Calculation:
   Conclusion:

Test the regression coefficient of above example:

Multiple Regression

Models the relationship between two or more independent variables and one dependent variable.

Equation: $Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \varepsilon$

Example: Predicting house price (Y) based on size, location, and age of the house.

Key Assumptions

For regression results to be valid and reliable, these assumptions must hold:

#	Assumption	Description
1	Linearity	Relationship between X and Y must be linear (Scatter plot should be linear)
2	Independence	Observations must be independent of each other (no geometric series) X: 5, 15,25,35,------ ➔ dependent
3	Homoscedasticity	Constant variance of error terms across all levels of X (each point have same variation) Observed – predicted= E➔ Var(E)= σ^2
4	Normality of Errors	Residuals (errors) should be normally distributed (bell shape)
5	No Multicollinearity	*(Multiple regression only)* Independent variables should not be highly correlated with each other (students consumption depends on parent's income & student's pocket money, Dependent variable: Students Consumptions; Independent are parent's Income & student's pocket money where as students pocket money is also dependent on parent's income)
6	No Autocorrelation	Error terms should not be correlated with each other (important in time series) E= 15,20,25,30,------